

Loading Data into R and Data Wrangling

Introduction to Lab

The objectives of this lab is to learn the basics about using R. You will learn the following in this lab:

1. How to load datasets into R Notebooks as a dataframe using the `read.csv()` command
2. How to view loaded datasets and use the `subset()` command to select a subset of a data frame
3. How to use the `summary()` command and basic functions to summarize data
4. How to use the aggregate command to get basic statistics from data frames
5. How to import data from qualtrics using the `qualtrics` package

This is possibly the most important lab all semester.

Seventy-five percent of the questions I get about R involve putting data into R and getting the data to load correctly. This lab covers this valuable skill. It will seem repetitive but this lab is very important, since every other lab will require you to load data from outside sources.

As mentioned in the readings and in the last lab, it's hard to put data into R manually, like what you did in the last lab assignment with the grades.

Normally, we will input data from other sources. I use a spreadsheet program to enter data and then import the completed data into R. As we covered in class, there is a very important way to enter data into R. A spreadsheet should be set up so that variables in columns and each observation as a row. The first row will contain the column names and each of the other rows will contain the data.

Getting Set Up

You can read more about this in Lab 1 regarding how to set up a R notebook and the folders on your computer or on RStudio on Posit cloud

The first thing you should do is create a folder on your computer for this lab, in the folder you have for this class. If you are on Posit Cloud, create a new folder using the New Folder button in RStudio in the bottom right window. After you do this, create an R notebook which contains this lab. Go ahead and delete the text that is there, change the title to something more appropriate, like “Lab 2”, and save this as Lab2.RMD in that folder.

Now, download the two data files for this lab. This data should be available on Canvas. The first datafile is lab2score.csv. The second is lab2_qualtrics.csv file.

Here is how to download the files. In Canvas go to the Files tab on the right and find the data files and select the three vertical dots to the right and click Download. This will pop up a window in your browser. Select Save or Save As and save this file to the folder you made for this lab.

Posit cloud If you're using Posit Cloud, you then need to upload the folder from your computer or iPad to the Cloud. When you download the file using the same instructions as before, save it anywhere on your computer. The desktop or downloads folder are good ideas. Then in Posit Cloud go to the folder that has your RStudio notebook and select the “upload” button on the top of the bottom right pane.

The most important part is to make sure your R notebook and your data files are in the same folder. If this does not happen, you will get an error that says:

```
Error in file(file, "rt") : cannot open the connection
```

When you see this error, that means you do not have the data in the same file folder, or you mistyped the name of the data frame.

Important One error a lot of people make is to move the RStudio notebook file after making it. You should never move that file from one folder to another using Finder or Windows explorer while it is open. If you need to move it, use Save As to save it in the correct folder.

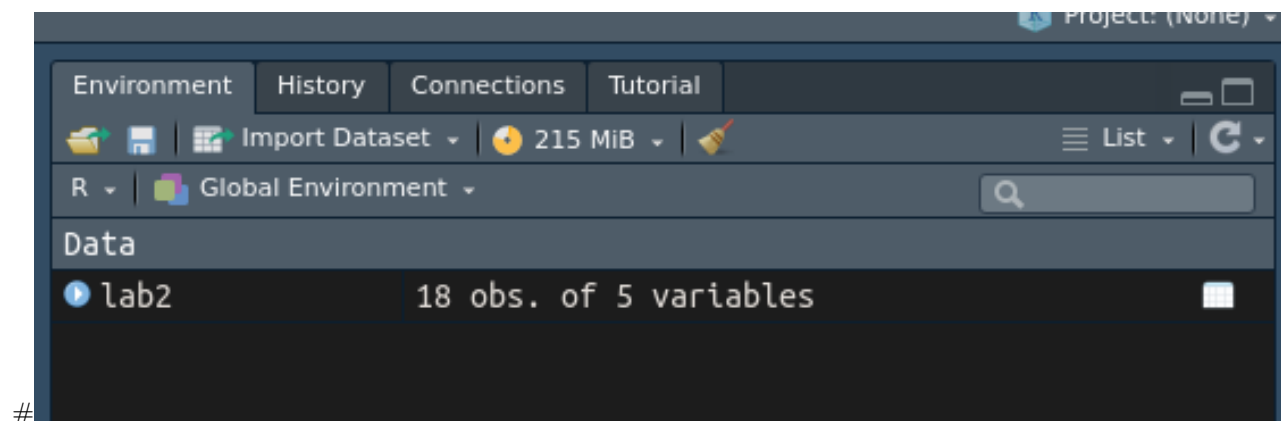
Now we will create a code chunk which loads the data into R. The code chunk will use the `read.csv` command and it will save the data as a data frame named `lab2`. Type the following and run the code chunk:

```
lab2 = read.csv('lab2score.csv')
```

When you start to type the file name, you can hit the tab button and select the file name from the drop-down list. For example, once you've typed: `lab2 = read.csv('l`, then you can hit tab and it will fill in the rest of the options for you.

Note that you have to type the file name exactly correct for R to load the data. If you misspell it, you will get the error that happens above. One common misspelling is to type `cvs` instead of `csv`.

If you loaded the data correctly, you should see a `lab2` dataframe in your environment. It will look like this:



Click on the icon to the right of “5 variables” to view that dataframe.

Whenever I give you data to enter into R, I will also give you a key that tells you about each of the variables. When you make your own data (for a research methods project) it is a good idea to make your own key for when you share it with other people. Here is the key below.

Variables: * Person: person's name * Education: highest level of education completed: none, HS * for high school, college for college, and graduate for graduate work * Sex: F for female, M for male * Test: type of test: Math or English * Score: score on a 50 question test

Whenever we create a new object, we may want to check what kind of object it is. We can use the `class()` command to do so. Type the following:

```
class(lab2)
```

```
## [1] "data.frame"
```

This tells us that `lab2` is a data frame. So we know we input the data correctly.

If we look at `lab 2` in the environment, we can also see the number of variables and observations.

1. How many observations are in x? How many variables? Please include this information in your annotation in your R Notebook.

Filtering rows for specific observations

In many data analyses, we will often want to filter our data to only have some rows from our main dataset. For instance, we may want to remove outliers. Or we might want to only look at one group within our data.

R has the `subset()` command to do this. `Subset` takes two arguments or inputs. The first is a data frame. The second is a **conditional**. A conditional is a comparison that gives a TRUE/FALSE answer. For `subset`, we have to give a column and compare it to a value.

For example, if I had a dataframe `x` and wanted to select all the values of `xvariable` that equaled 10, I would type `subset(x, xvariable == 10)`.

Try the following. In your annotation, write what part of the data frame is being selected here.

5a.

```
subset(lab2, lab2$Education == "College")
```

5b.

```
subset(lab2, lab2$Score > 20)
```

5c.

```
subset(lab2, lab2$Name != "Alex")
```

The last example is very important to remember because occasionally we might want to remove a single value, such as an outlier. This is how we would do so.

For R, you can compare two values with a statement called a conditional. The conditionals in R are `==` (two equals signs, which tests whether two variables are equal or if each element of a variable is equal to a certain value), `>` and `<` (doing greater than or lesser than), and `!=` which tests not equal. When doing a conditional, R returns TRUE or FALSE.

Try typing the following. What happens?

```
lab2$Sex == "F"
```

Now use code chunks to answer the next two questions. Please include in your annotation which question you are answering.

6. Using the `subset` command, what command would you type to create a new dataframe with only rows that contained people who had a High School education?

7. What command would you use to have only rows for the English test?

8. Based on what you learned above, how would you remove Bob's data from the dataframe.

Summarizing data

One easy way to look at the summary of a dataframe is to use the `summary()` command. This command gives the summary of an object, no matter what kind of object it is.

Go ahead and type the following:

```
summary(lab2)
```

This gives us a summary of each of the variables including the minimum, maximum, mean, and median. This can be helpful to view what is happening in a data frame and reporting basic statistics. Another way of doing this is by using the `dataframe$variable` notation and using a basic command like `mean()` or `sd()`

For example, to get the mean of Score, I would type:

```
mean(lab2$Score)
```

```
## [1] 26.94444
```

9. How would I get the standard deviation of score?

Using aggregate to get statistics separated by a grouping variable

Another really important way to analyze data is to compare groups. In the `lab2` dataframe, note that only one variable is actually containing numbers. All the other variables tell us about groups. For instance, the `Person` variable tells us what person's score it is. The `Sex` variable tells us the sex of the person.

We may want to look at summary statistics broken up by group. For instance, we may want to compare Math versus English scores. The `aggregate` command allows us to do so. The `aggregate` command is complicated and has several different parts. Here is the command that would give us the mean for Score, broken up by Test.

```
aggregate(Score~Test, lab2, FUN = mean)
```

```
##      Test      Score
## 1 English 28.00000
## 2   Math 25.88889
```

The `aggregate` command has three parts. The first part is a **formula** which is a way of breaking up the data. The second part is the dataframe we are looking at. The third part is the function we are applying. It has nothing to do with fun, as we know there's no fun in statistics.

The formula has two parts, divided by a tilde (~) which is the key to the left of the 1 key on your keyboard. The first part of the formula is the dependent variable. This is the numeric variable that we want to analyze. The second part is the grouping variable. This is the variable which we use to determine the groups. In this case, we use Score as the dependent variable and Test as the grouping variable.

If we wanted to compare females to males, we would change the grouping variable from Test to Sex. Type the following:

```
aggregate(Score~Sex, lab2, FUN = mean)
```

What results do you get?

If we want to change the type of function, we can change the `FUN = mean` part to some other command. If we really want to raise the fun, we might look at a median instead of a mean.

```
aggregate(Score~Sex, lab2, FUN = median)
```

Using this logic and aggregate, create code chunks to do the following:

10. Get the mean score for each person. This will involve using Person as your grouping variable.
11. Get the standard deviation for each test (English versus Math).
12. Get the mean and median for Score broken up by Education level. Is this what you would predict?.

Importing data from Qualtrics

Qualtrics data is exported as CSVs, but because Qualtrics does not like to play nicely with others, we can't import them straight into R. Some amazing people wrote a package that will help us do that.

In Qualtrics, you can export your data as a CSV file in the “Data and Analysis” section. After doing that, you will have a data file. As part of this assignment, I had you download an exported Qualtrics file called `lab2_qualtrics.csv`.

To import this, first we have to install the `qualtRics` package. Note only the R is capitalized in this word (so cute). Install this package by going to Tools -> Install Packages at the top (or the Packages tab on the side). Type in `qualtRics` and then you will install the package.

Then load the package by creating a code chunk that says:

```
library(qualtRics)
lab2.qualtrics = read_survey('lab2_qualtrics.csv')
```

You'll probably get a long output in your R notebook which tells you how RStudio inputted the variables. Click on the `lab2.qualtrics` dataframe to see if it inputted correctly. There should be 69 observations of 72 variables. That's a lot of data, and much of it is probably nonsense.

We're not going to look at this now, but we will use this in future assignments where you collect your own data!

Summary

Now go ahead and knit your document. You may have errors when you try to knit it. Find the line number and try to fix your errors. If you have an error with the `read.csv()` command, it may be because your data frame file name has a typo or the data and R notebook are saved in different file folders. Once you knit this assignment, go ahead and upload it and you're done! Great job. You're well on your way to being a world-class data analyst. But before you get too excited, you may want to make sure you know how to do the following, which we discussed in this lab:

- Load a dataframe into R using the `read.csv()` command
- Select a subset of the data frame using the `subset()` command
- Use the `summary()` command to get an overview of a data frame
- Use the `aggregate()` command to get the mean broken down by a variable
- Use the `qualtrics` package to load survey data downloaded from Qualtrics